

Import Data using Transform Maps (Spreadsheet)

ServiceNow Project Documentation

Phase 6: Project Documentation

Item	Details
Project Title	Import Data using Transform Maps (Spreadsheet)
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Platform	ServiceNow
Team ID	SWTID-2026-2373

6.1 Project Overview

This micro project imports structured employee data from an external spreadsheet into the ServiceNow platform using Import Sets and Transform Maps. It simulates a real-world scenario in which bulk employee data arrives in Excel format and must be migrated into ServiceNow accurately and efficiently.

The spreadsheet is first loaded into an Import Set table, which acts as a staging area. A Transform Map then maps the source fields to the target table fields. When the transformation runs, records are created in the target table and validated. Coalesce is later enabled to prevent duplicate records, and the data is presented through reports and a dashboard.

6.2 Objectives

- Load employee data from an Excel file into a ServiceNow staging table.
- Create a custom target table and map source fields to it using a Transform Map.
- Execute the transformation and verify the created records.
- Prevent duplicate records using Coalesce, and handle inserts and updates on repeated imports.
- Build reports and a dashboard for employee data visibility.

6.3 Tools and Components Used

Component	Details
Platform	ServiceNow (Global application scope)
Source file	Google Sheets, downloaded as Sample Spreadsheet.xlsx
Custom table	Employee Test (u_employee_test)
Staging table	Employee Import (u_employee_import)
Transform Map	Sample Spreadsheet Import

Component	Details
Reports	Employees by Department (Pie), Employees by Location (Bar), Employee List Report (List)
Dashboard	Employee Analytics Dashboard (created through PA_DASHBOARDS.FORM)

6.4 Implementation Summary

Phase	Activity	Navigation / Action	Outcome
1	Prepare the sheets	Create a Google Sheet with sample data, then File → Download → Microsoft Excel (.xlsx)	Sample Spreadsheet.xlsx
2	Create custom table	Tables → Create New; add 5 String fields through Form Layout	u_employee_test with 5 fields
3	Import Set table	Load Data → Create table, source File, sheet 1, header row 1	u_employee_import; 15 records processed
4	Create Transform Map	Create Transform Map → Auto Map Matching Fields / Mapping Assist → Transform	Records created in Employee Test
5	Transform and validate	Open Employee Tests; arrange columns with Personalize List Columns	15 of 15 records verified
6	Enable Coalesce	Transform Map → Field Maps → set Coalesce true on u_employee_id	Duplicate control enabled
7	Import new data	Load Data → Existing table; Run Transform; check Transform History	2 inserts, 2 updates; re-import ignored
8	Create reports	Reports → New; create 3 reports on Employee Test	Pie, Bar and List reports
9	Dashboard	Create dashboard; Share → Add to Dashboard for each report	Employee Analytics Dashboard

6.5 Configuration Details

Custom table: Employee Test

Column label	Column name	Type
Employee ID	u_employee_id	String
Employee Name	u_employee_name	String

Column label	Column name	Type
Email	u_email	String
Department	u_department	String
Location	u_location	String

Import Set table (Load Data)

Setting	Value
Import set table	Create table
Label / Name	Employee Import / u_employee_import (auto populated)
Source of the import	File (Sample Spreadsheet.xlsx)
Sheet number / Header row	1 / 1
Result	State: Complete, Completion code: Success (Processed: 15, inserts 15, errors 0)

Transform Map

Setting	Value
Name	Sample Spreadsheet Import
Source table	Employee Import [u_employee_import]
Target table	Employee Test [u_employee_test]
Mapping method	Auto Map Matching Fields, checked with Mapping Assist
Execution	Transform link → select the import set → Transform button

Source field	Target field	Coalesce
u_employee_id	u_employee_id	true
u_name	u_employee_name	false
u_email	u_email	false
u_department	u_department	false
u_location	u_location	false

Reports and dashboard

Report	Type	Configuration	Style
Employees by Department	Pie	Group by Department, Aggregation Count	Default color palette
Employees by Location	Bar	Group by Location, Aggregation Count	Use one color
Employee List Report	List	Columns: Employee ID, Name, Email, Department, Location	No changes needed

All three reports were added to the Employee Analytics Dashboard using Share → Add to Dashboard. Further widgets can be added from the + icon, the background color can be configured, and access can be restricted by users, groups and roles from the Share option.

6.6 Results and Validation

Import scenarios

Scenario	Total	Inserts	Updates	Ignored
First import	15	15	0	0
Second import (coalesce on Employee ID)	4	2	2	0
Same sheet imported again	4	0	0	4

Data used in the second import

Employee ID	Name	Email	Department	Location	Result
SB-0004	Ajay	test4@gmail.com	ServiceNow	Hyderabad	Updated (name changed from Ajay Kumar)
SB-0010	Raviteja	test18@gmail.com	ServiceNow	Hyderabad	Updated (email changed from test10@gmail.com)
SB-0016	Siri	test16@gmail.com	Aiml	Chennai	Inserted
SB-0017	Naga sai	test17@gmail.com	Aiml	Hyderabad	Inserted

Final state of the Employee Test table

- Total records: 17.
- Department distribution: ServiceNow 11 (64.71%), Salesforce 4 (23.53%), Aiml 2 (11.76%).
- Location distribution: Hyderabad 16, Chennai 1.
- Every import and transform finished with State Complete and Completion code Success.

6.7 Issues and Resolutions

Issue	Resolution
Duplicate records on repeated imports	Enabled Coalesce on u_employee_id in the Transform Map. Existing IDs are updated, and identical rows are ignored.
Dashboard form did not open by its backend name (PA_DASHBOARDS.FORM)	In Access Control (ACL), opened the pa_dashboards create operation and set Admin overrides to true, then updated the record.
Columns appeared in a different order in the list view	Rearranged them with Personalize List Columns (Employee ID, Name, Email, Department, Location).
Second import needed the same Transform Map	Loaded the file into the existing table (Employee Import) and used Run Transform, which pre-selects the Sample Spreadsheet Import map.

6.8 Conclusion

1. The project implemented an employee data import process in ServiceNow using Import Sets and Transform Maps, with accurate mapping of source fields to the target table.
2. Enabling Coalesce on Employee ID lets the system recognise existing employees on every import. Repeated imports update existing records instead of creating duplicates, which keeps the employee table clean and consistent.
3. Reports and a dashboard give administrators and HR teams a single view of department and location distribution and the full employee list.

6.9 Suggested Future Enhancements

- Add reports on Transform History to track import inserts, updates and error counts.
- Schedule imports with a Data Source instead of uploading files manually.
- Add validation through transform scripts, for example checking email format.
- Restrict the dashboard to HR roles through the Share option.